

STANDARD OPERATING PROCEDURE

SOP-MNT-001

Centrifugal Pump Mechanical Seal Replacement and Precision Alignment Procedure

Applicable Equipment:	P-101, P-102, P-103, P-104 (API 610 centrifugal pumps)
Revision:	Rev. 4 (Effective January 10, 2025)
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1. Purpose and Scope

This procedure defines the step-by-step process for mechanical seal replacement and precision shaft alignment on API 610 centrifugal pumps within the Petromax Baytown Complex. It applies specifically to the following equipment: P-101 (Centrifugal Feed Pump), P-102 (Booster Pump), P-103 (Transfer Pump), and P-104 (Recirculation Pump). All personnel performing this work must hold a minimum of Maintenance Technician Level II certification and have completed the Petromax Rotating Equipment Safety Training (Course RE-401).

2. Safety Precautions

DANGER: These pumps handle crude oil and hydrocarbon products at elevated temperatures (up to 350 degF) and pressures (up to 400 psig). Failure to follow proper isolation and depressurization procedures can result in serious injury or death.

Required PPE:

- Flame-resistant coveralls (FRC) per NFPA 2112
- Safety glasses with side shields (ANSI Z87.1)
- Chemical-resistant gloves (neoprene or nitrile)
- Steel-toe boots (ASTM F2413)
- Hard hat (ANSI Z89.1 Type I)
- H2S personal monitor (set to 10 ppm alarm / 20 ppm STEL)

Required Permits:

- Hot Work Permit (if any grinding, welding, or cutting is required)
- Confined Space Entry Permit (if entering pump enclosure below grade)
- Lock-Out/Tag-Out (LOTO) per OSHA 29 CFR 1910.147
- Line Break Permit (for opening process connections)

3. Pre-Job Planning

3.1 Review the most recent vibration analysis report for the pump. For P-101, reference the baseline vibration data from the August 2018 commissioning report (VIB-P101-BASELINE). Current vibration trending data is available in the plant condition monitoring system (Emerson AMS Machinery Manager).

3.2 Verify mechanical seal part number against the equipment bill of materials:

- P-101: John Crane Type 4620, API Plan 53B (pressurized dual seal)
- P-102: Flowserve QBW, API Plan 11 (single seal with flush)
- P-103: John Crane Type 4620, API Plan 53B
- P-104: Burgmann M7N, API Plan 21 (single seal with cooler)

3.3 Ensure the following special tools are available:

- Laser alignment system (Pruftechnik ROTALIGN Ultra iS)
- Dial indicator set with magnetic base
- Bearing heater (SKF TIH 030m)
- Torque wrench set (calibrated within last 6 months)
- Seal installation sleeve (pump-model specific)

4. Isolation and Preparation

4.1 Notify the Control Room Operator (CRO) of the impending pump isolation. Coordinate with Operations to transfer service to the designated spare pump (if available).

4.2 Implement LOTO per the pump-specific LOTO procedure:

- P-101 LOTO: LOTO-P101-001 (3 energy isolation points: motor breaker, suction valve, discharge valve)
- P-102 LOTO: LOTO-P102-001 (4 energy isolation points: motor breaker, suction valve, discharge valve, minimum flow recirculation valve)

4.3 Drain the pump casing and flush with nitrogen to remove residual hydrocarbons. Verify atmosphere with a 4-gas detector (LEL < 10%, O₂ 19.5-23.5%, H₂S < 10 ppm, CO < 25 ppm).

4.4 Disconnect coupling guard. Remove coupling spacer (record spacer shim pack thickness for reinstallation reference).

5. Mechanical Seal Removal

5.1 Remove the gland plate bolts (typically 8 x 5/8" UNC, Grade 8). Note: for P-101, the gland uses Superbolt multi-jackbolt tensioners -- refer to the Superbolt tightening procedure (torque each jackbolt to 45 ft-lbs in a star pattern, 3 passes).

5.2 Slide the gland plate along the shaft toward the coupling end. For dual seals (P-101, P-103), first drain the barrier fluid reservoir and depressurize the seal chamber.

5.3 Using the seal installation/removal sleeve, carefully slide the seal assembly off the shaft. Inspect the shaft sleeve for scoring, wear, or corrosion. Shaft sleeve OD should measure within 0.001" of nominal (reference pump data sheet for exact dimension).

5.4 Inspect the seal chamber bore for scoring or deposits. Clean with Scotch-Brite and solvent. Measure seal chamber bore ID -- must be within 0.002" of nominal.

5.5 Document all inspection measurements on Form MNT-SEAL-001 (Seal Replacement Inspection Checklist) and photograph the removed seal faces for failure analysis records.

6. Mechanical Seal Installation

6.1 Verify the new seal part number matches the equipment BOM. Check the seal setting height against the seal manufacturer's installation drawing.

6.2 Lubricate the shaft sleeve and seal O-rings with a compatible lubricant (for hydrocarbon service, use Krytox GPL-105). DO NOT use petroleum-based lubricants on elastomers.

6.3 Install the seal using the installation sleeve. For John Crane Type 4620 (P-101, P-103), the seal setting clips must be in place during installation -- remove only after the gland plate is torqued to specification.

6.4 Torque the gland plate bolts to the values specified in the seal installation drawing:

- P-101/P-103 (Superbolt): 45 ft-lbs per jackbolt
- P-102 (standard gland): 35 ft-lbs
- P-104 (standard gland): 30 ft-lbs

6.5 For dual seal systems (P-101, P-103), fill the barrier fluid reservoir with the specified fluid (Petromax spec: synthetic barrier fluid, Shell Thermia B or equivalent). Pressurize the barrier system to 25 psi above seal chamber pressure.

7. Precision Shaft Alignment

7.1 Install the laser alignment equipment (Pruftechnik ROTALIGN Ultra iS) per manufacturer's instructions. Mount sensor heads on the pump shaft and motor shaft coupling hubs.

7.2 Acceptable alignment tolerances (at operating temperature):

- Offset (Parallel): 0.002" maximum
- Angular: 0.0005"/inch maximum

7.3 Perform initial alignment measurement. Record values:

- Horizontal offset
- Vertical offset
- Horizontal angularity
- Vertical angularity

7.4 Make corrections by shimming the motor feet (for vertical corrections) and sliding the motor (for horizontal corrections). Use stainless steel pre-cut shim stock only -- minimum shim thickness 0.002", maximum shim pack 4 shims per foot.

7.5 Re-measure and iterate until alignment is within tolerance at all positions. Document final alignment values on Form MNT-ALIGN-001 (Precision Alignment Record).

7.6 Apply thermal growth compensation as specified in the pump data sheet. For P-101, the calculated thermal offset is 0.005" (motor high) to compensate for operating temperature differential between pump (285 degF) and motor (180 degF ambient).

8. Reassembly and Startup

8.1 Reinstall coupling spacer and guard. Verify coupling guard clearances per plant standard.

8.2 Remove all LOTO locks and tags per the de-isolation procedure. Verify with the CRO that all isolation points have been restored.

8.3 Slowly open the suction valve and verify pump casing is fully vented (open vent valve on top of casing until steady liquid stream is observed).

8.4 Coordinate with CRO for initial pump start. Monitor for:

- Seal leakage (visual check at gland plate)
- Vibration (handheld readings at all bearing housings)
- Bearing temperature (infrared gun readings)
- Discharge pressure and flow rate

8.5 After 30 minutes of stable operation, take a full set of vibration readings (triaxial at each bearing housing) and compare to the pre-job baseline.

8.6 After 24 hours of operation, perform a follow-up check of seal leakage, vibration levels, and bearing temperatures. Enter results in the Computerized Maintenance Management System (IBM Maximo, Work Order reference).

8.7 Schedule a 30-day post-maintenance vibration survey to verify sustained acceptable operation.

9. Documentation and Records

All work performed under this SOP must be documented in the following:

1. IBM Maximo Work Order -- include labor hours, parts consumed, and condition codes
2. Form MNT-SEAL-001 -- Seal Replacement Inspection Checklist
3. Form MNT-ALIGN-001 -- Precision Alignment Record
4. Vibration baseline file uploaded to Emerson AMS Machinery Manager
5. Photographic record of removed seal faces (stored in Equipment History folder)

Completed forms must be submitted to the Maintenance Planner within 48 hours of job completion. Any deviations from this procedure must be documented on a Management of Change (MOC) form and approved by the Reliability Engineer prior to implementation.